25900141 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCMBG101 - MICROBIAL PHYSIOLOGY AND FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. What is the term for initial phase of a batch culture? [K] / [CO1]

   a). Decline phase b). Lag phase

   c). Log phase d). Death phase

2. Choose the term used for a medium that support the growth of a wide range of microorganisms. [K] / [CO2]

   a). Selective medium b). Differential medium

   c). Enrichment medium d). General purpose medium

3. Which of the following is not a component of nutrient broth? [K] / [CO2]

   a). NaCl b). Peptone

   c). Yeast extract d). Ammonia

4. Choose the term for a medium that is used to differentiate between different types of microorganisms. [K] / [CO2]

a). Selective medium b). Differentilal medium

c). General purpose medium d). All of the above

5. Select the term for bacteria that uses oxygen for growth. [K] / [CO3]

a). Aerobes b). Anaerobes

c). Thermophiles d). Mesophiles

6. Which of the following processes lead to the production of alcohol from microbial action on carbohydrates? [K] / [CO3]

a). Fermentation b). Hydrolysis

c). Oxidation d). Reduction

7. Which of the following is not an example for biological structure of food that influence microbial growth? [K] / [CO3]

a). Nutrient content b). Shell of eggs

c). Shell of nuts d). Skin of fruits

8. Choose the pathogen that is spread through raw milk. [U] / [CO4]

a). Salmonella b). Listeria

c). E. coli d). all of the above

9. Compare the forms of microorganisms listed below and select the one that has the greatest resistance to UV radiation. [An] / [CO4]

a). Bacterial endospore b). Vegetative bacteria

c). Fungal spores d). Yeast cells

10. What is the primary antimicrobial agent found in wood smoke? [U] / [CO4]

a). Alcohol b). Formaldehyde

c). Antibiotics d). Acids

[1x10 = 10]

Part B

Fill In The Blanks

Answer all questions

Each question carries **1** marks

11. Most of the bacteria multiply by _____. [U] / [CO1]

12. _____ is an example of selective media. [K] / [CO2]

13. _____ bacteriological media contain known chemical compounds. [U] / [CO2]

14. _____ is an intrinsic factor that influence microbial growth in foods. [U] / [CO3]

15. …… is an example of a chemical hurdle. [U] / [CO4]

[1x5 = 5]

Part C

State Whether True or False

Answer all questions

Each question carries **1** marks

16. Lag phase is the last phase of bacterial growth curve. [U] / [CO1]

17. Mac Conkey agar is not a selective medium. [U] / [CO2]

18. Pre-harvest contamination from natural flora of plants and animals is always harmless and does not pose any risk to food safety. [U] / [CO3]

19. Sexual spores in fungi are formed through the fusion of male and female gametes. [U] / [CO3]

20. Dried foods are completely free of contaminants and spoilage organisms. [U] / [CO4]

[1x5 = 5]

Part D

Very Short Answer Type Questions

Answer any **10** questions

Each question carries **1** marks

21. What are photoheterotrophs? [K] / [CO1]

22. What is Mac Conkey agar? [K] / [CO2]

23. Name a differential medium. [K] / [CO2]

24. What is nutrient agar? [K] / [CO2]

25. What are anaerobes? [K] / [CO2]

26. What is spoilage? [K] / [CO3]

27. Name the binomial of distiller's yeast. [K] / [CO3]

28. Indicate the major kinds of contaminants that enter into food from workers. [U] / [CO3]

29. Expand GRAS. [K] / [CO4]

30. Give two examples of natural asepsis in foods. [U] / [CO4]

31. Indicate the applications of ethylene oxide in food preservation. [U] / [CO4]

32. List out the advantages of solar drying. [K] / [CO4]

[1x10 = 10]

Part E

Short Essay Questions

Answer any **4** questions

Each question carries **3** marks

33. Classify bacteria based on their PH requirements. [U] / [CO1]

34. Explain spread plate method. [U] / [CO2]

35. Discuss about the characteristics that make Pseudomonas important in foods. [U] / [CO3]

36. Compare perishable and semi-perishable foods. Give examples. [U] / [CO3]

37. Discuss about any four advantages of using hurdle technology. [U] / [CO4]

38. List out the steps involved in the canning of vegetables. [U] / [CO4]

[3x4 = 12]

Part F

Long Essay Question

Answer any **1** questions

Each question carries **8** marks

39. Explain the basic media components. Classify media based on their chemical nature. [U] / [CO2]

40. Discuss in brief about the different types of drying. [U] / [CO4]

[8x1 = 8]